

Machine Perception and the Limits of Representation

Jane Q. Student

Essay for PHIL 210, Philosophy of Mind

Whether a machine can be said to perceive depends on what we take representation to be. Layered statistical models now label images and transcribe speech with striking accuracy, and some writers take this as evidence that such systems form representations of the world.¹ This essay argues that accuracy alone settles nothing about representation, because the same behaviour is compatible with several accounts of what, if anything, the system represents.

¹ Y. LeCun, Y. Bengio and G. Hinton, Deep learning, Nature 521 (2015) 436-444, <https://doi.org/10.1038/nature14539>.

The argument of the previous section applies to any system trained only on labelled examples.

References

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436-444.
<https://doi.org/10.1038/nature14539>